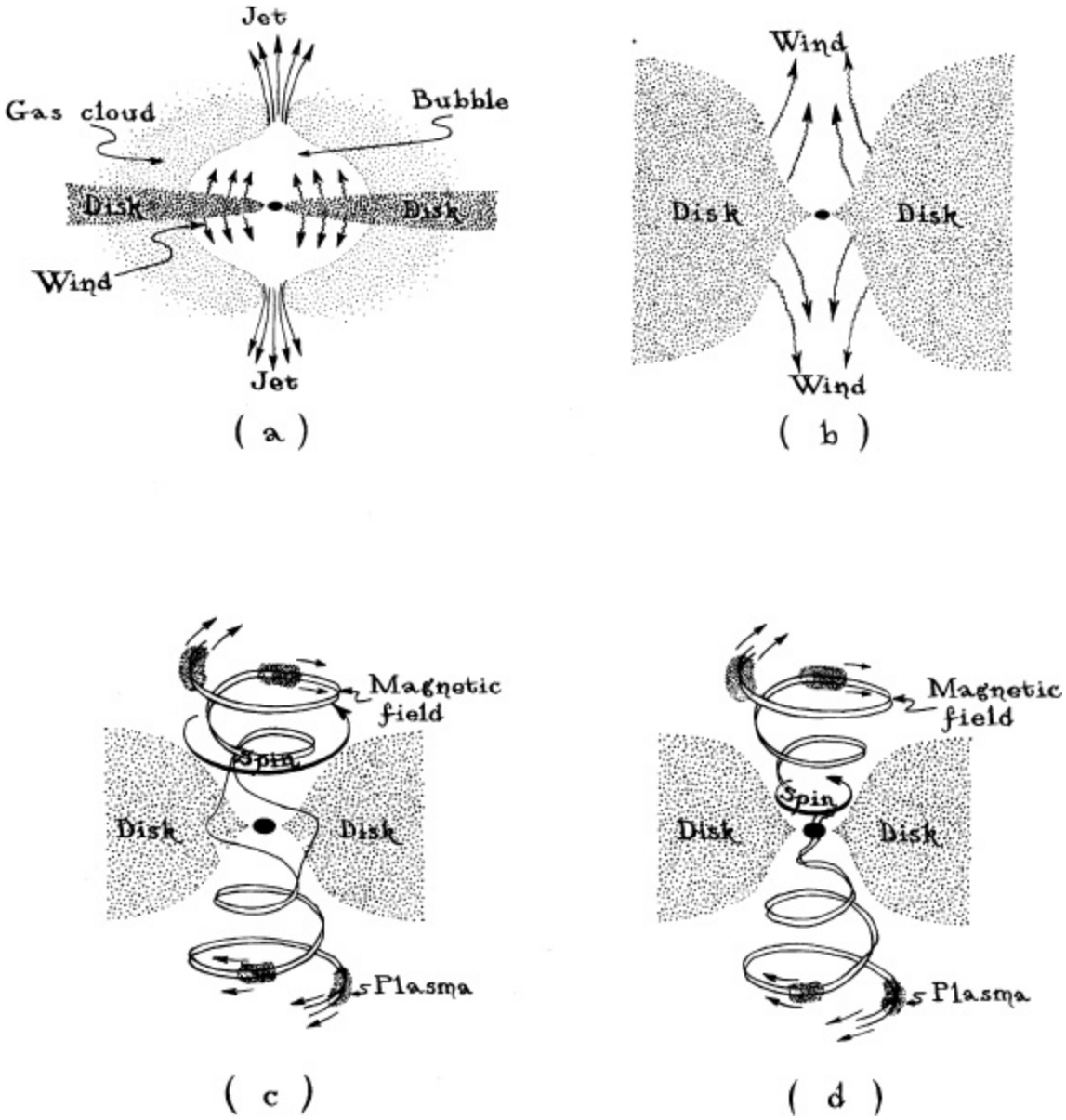


9.7 Four methods by which a black hole or its accretion disk could power twin jets. (a) A wind from the disk blows a bubble in a surrounding, spinning gas cloud; the bubble's hot gas punches orifices through the cloud, along its spin axis; and jets of hot gas shoot out the orifices. (b) The disk is puffed up by the pressure of its great internal heat, and the surface of the puffed, rotating disk forms two funnels that collimate the disk's wind into two jets. (c) Magnetic field lines anchored in the disk are forced to spin by the disk's orbital rotation; as they spin, the field lines fling plasma upward and downward, and the plasma, sliding along the field lines, forms two magnetized jets. (d) Magnetic field lines threading through the black hole are forced to spin by the swirl of the hole's space, and as they spin, the field lines fling plasma upward and downward to form two magnetized jets.

The fourth method of producing jets is more interesting than the others and requires more explanation. In this fourth method, the hole is threaded by magnetic field lines as shown in Figure 9.7d. As the hole spins, it drags